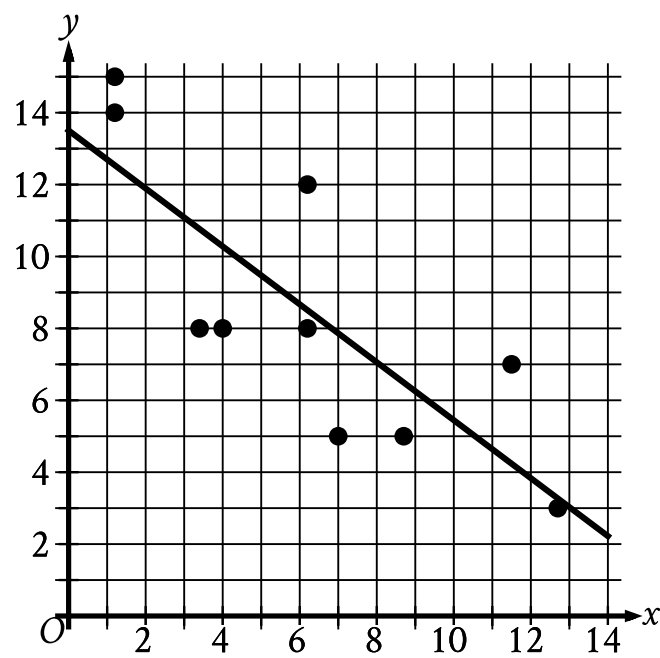


Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	Easy

Question

The scatterplot shows the relationship between two variables, x and y . A line of best fit is also shown.



Which of the following equations best represents the line of best fit shown?

Answer

- A. $y = 13.5 + 0.8x$
- B. $y = 13.5 - 0.8x$
- C. $y = -13.5 + 0.8x$
- D. $y = -13.5 - 0.8x$

Correct Answer: B

Rationale

Choice B is correct. The line of best fit shown intersects the y-axis at a positive y-value and has a negative slope. The graph of an equation of the form $y = a + bx$, where a and b are constants, intersects the y-axis at a y-value of a and has a slope of b . Of the given choices, only choice B represents a line that intersects the y-axis at a positive y-value, **13.5**, and has a negative slope, **-0.8**.

Choice A is incorrect. This equation represents a line that has a positive slope, not a negative slope.

Choice C is incorrect. This equation represents a line that intersects the y-axis at a negative y-value, not a positive y-value, and has a positive slope, not a negative slope.

Choice D is incorrect. This equation represents a line that intersects the y -axis at a negative y -value, not a positive y -value.